Python File: 01_trace_control_flow.py

```
x=7
if x>5:
    print("High")
elif x>2:
    print("medium")
else:
    print ("lOW")

total=0
for i in range(1,5):
    total+=i
print("total")
count=0
while count<3:
    print("loop:",count)
count+=1
```

Python File: 02_voting.py

```
age=int(input("Enter your age "))
if age<18:
    print("You are not eligible for vote")
else:
    print("You are eligible for vote")
```

Python File: 03__bmi_cassification.py

```
weight=float(input("Enter weight kg: "))
height=float(input("Enter height in meters: "))
BMI = weight / (height ** 2)
print(BMI)
if  BMI<18.5:
    print("Underweight")
elif BMI<25:
    print("Normal")
elif BMI<30:
    print("Overweight")
else:
    print("Obese")
```

Python File: 05_multiplication_table.py

```
for i in range(1,11):
    print("5 X ",i,"=",5*i)
```

Python File: 06_break_continue.py

```
for i in range(1,20):
    if  i%3==0:
        continue
    if  i==17:
        break
    print(i)
```

Python File: 07_patterns.py

```
for i in range(1, 6):
    print("*" * i)

for i in range(5, 0, -1):
    print("*" * i)
```

Python File: 08_comprehensive.py

```
values = []

for number in range(1, 6):
    values.append(number ** 2)

values_comp = [number ** 2 for number in range(1, 6)]

print(values)
print(values_comp)
even_numbers = [number for number in range(1, 51) if number % 2 == 0]

odd_numbers = [number for number in range(1, 51) if number % 2 != 0]

print("Even:", even_numbers)
print("Odd:", odd_numbers)
```

Python File: 09_marks_anylsis.py

```
marks = [78, 65, 92, 85, 48, 91]

passing = []
failing = []

for mark in marks:
    if mark >= 50:
        passing.append(mark)
    else:
        failing.append(mark)

total = 0

for mark in marks:
    total += mark

average = total / len(marks)

highest = max(marks)
lowest = min(marks)

print("Passing:", passing)
print("Failing:", failing)
print("Average:", average)
print("Highest:", highest)
print("Lowest:", lowest)
```

Python File: 10_debugging.py

```
# G1. Missing colon

# Error: Missing colon (:) after the for loop condition.
# Cause: Python requires a colon at the end of a for loop statement.
# Fix: Add a colon after range(5).

for i in range(5):
    print(i)


# G2. Assignment and comparison

# Error: Assignment operator (=) is used instead of comparison operator (==).
# Cause: = assigns a value, while == compares two values.
# Fix: Use == to compare x with 5.

x = 5

if x == 5:
    print(x)


# G3. Infinite loop

# Error: The while loop is infinite.
# Cause: The value of x never changes, so x < 10 always remains True.
# Fix: Increase x inside the loop so the condition eventually becomes False.

x = 0

while x < 10:
    print(x)
    x = x + 1
```

Python File: 11_traffic_signal.py

```
# Part H - Traffic Signal Decision

# Read traffic density
density = input("Enter traffic density (low, medium, or high): ").lower()

# Read emergency vehicle status
emergency = input("Is an emergency vehicle present? (yes/no): ").lower()

# Decide green-light time based on traffic density
if density == "low":
    green_time = 20
elif density == "medium":
    green_time = 40
elif density == "high":
    green_time = 60
else:
    print("Invalid traffic density. Please enter low, medium, or high.")
    green_time = None

# Emergency override or normal decision
if green_time is not None:
    if emergency == "yes":
        print("Emergency override: route receives immediate green.")
    else:
        print("Normal green-light time:", green_time, "seconds")
```

Python File: tempCodeRunnerFile.py

```
values =[]
for number in range(1,6):
    values.append(number**2)
    values_comp=[number**2 for number in range (1,6)]
    print(values)
    print(values_comp)
```